Armaan Amatya

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Education

University of Houston Houston, TX
Bachelor of Science in Computer Science and Mathematics *Aug. 2024 – Dec. 2026*

Experience

Autonomize AI Austin, TX
Machine Learning Engineer Intern *May 2026 – Present*

- Building deep RL post-training algorithms with reward-signal design to improve LLM alignment in clinical workflows
- Deployed Qwen2.5-VL-32B on a 16GB T4 via 4-bit AWQ quantization and CPU offloading, cutting VRAM ∼4x
- Developed a visual token compression pipeline for multimodal OCR workflows, pruning 50%+ redundant image tokens to cut inference latency 35% and KV-cache memory 40% and raise throughput 1.5x at <2% OCR accuracy loss

University of Houston Houston, TX
Undergraduate Researcher *Jan. 2026 – Present*

- Cut multimodal LLM inference latency ∼40% via a novel token compression method that reduces visual tokens ∼60%, shortening attention context and KV-cache footprint at ∼98% of baseline accuracy on Qwen2.5/3-Omni
- Deployed compressed models to NVIDIA Jetson Orin for on-device inference after multi-GPU distributed training (PyTorch DDP), cutting VRAM ∼45% and sustaining ∼2x throughput (tokens/s); won Best Research Award

Artinafti Austin, TX
AI Engineer Intern *Jan. 2026 – May 2026*

- Improved visual fidelity 25% by fine-tuning diffusion-based upscaling and generation models with LoRA adapters and detail-preserving objectives for text and faces, adding photographic color-tone control
- Reduced inference latency 30% with scalable model-serving APIs (FastAPI, NestJS, Redis) and request batching

AutoHDR Austin, TX (Remote)
Machine Learning Engineer Intern *Jan. 2026 – Apr. 2026*

- Cut per-image turnaround from ∼5 minutes to under 15 seconds across 10,000+ listing photos monthly by building high-throughput diffusion pipelines (PyTorch) with queue-backed batch GPU inference
- Reduced redundant processing ∼25% with an image-deduplication model on contrastive CLIP embeddings
- Produced 1,000+ property tour videos monthly with a depth-aware image-to-video generation pipeline

FuseMachines New York, NY
Machine Learning Engineer Intern *May 2024 – Aug. 2024*

- Achieved 98% precision on real-world attendance tracking by fine-tuning a FaceNet model
- Served real-time inference to a production mobile app (FastAPI, Docker) with async batched requests

Projects

Mixed-Precision GEMM Kernels in Triton | *Triton, CUDA, PyTorch, Nsight Compute*

- Built and autotuned a tiled FP16 GEMM kernel in Triton, benchmarked vs. cuBLAS on RTX and RX GPUs
- Profiled occupancy, memory bandwidth, and kernel launch overhead in Nsight Compute to isolate bottlenecks

Real-Time Distributed Keyword Spotting | *PyTorch, Wav2Vec2, CUDA, VxWorks, UDP*

- Held the CUDA Wav2Vec2 stage of a three-node pipeline to 12.5 ms p99 against a 150 ms hard real-time deadline, benchmarked over 72 scheduler (RMS/EDF/LLF), precision and preemption configurations

OpenApplier | *Rust, AWS, Microservices, RAG*

- Engineered resilient ML serving infra handling 10K+ daily inference requests with container orchestration and real-time observability

Domain QA System for Scientific PDFs | *HuggingFace, PEFT, BitsAndBytes, FAISS, LangChain*

- Post-trained Mistral-7B with QLoRA (4-bit NF4 quantization) on 234 QA pairs; built hybrid RAG over a 75-paper FAISS corpus; shipped quantized GenAI inference on HuggingFace Spaces, scored by GPT-4o LLM-as-judge

Technical Skills

Languages: Python (async/concurrent), C++, CUDA, Triton, Rust, SQL, Bash
GPU & Inference: low-level kernel optimization, numerics validation, PyTorch, vLLM, TensorRT, quantization (4-bit AWQ/QLoRA), inference optimization (latency & throughput), request batching
AI/ML: LLMs & modern model architectures, transformers & attention, RL (PPO/GRPO/RLHF), post-training & alignment, end-to-end ML pipelines, RAG, Docker, Kubernetes